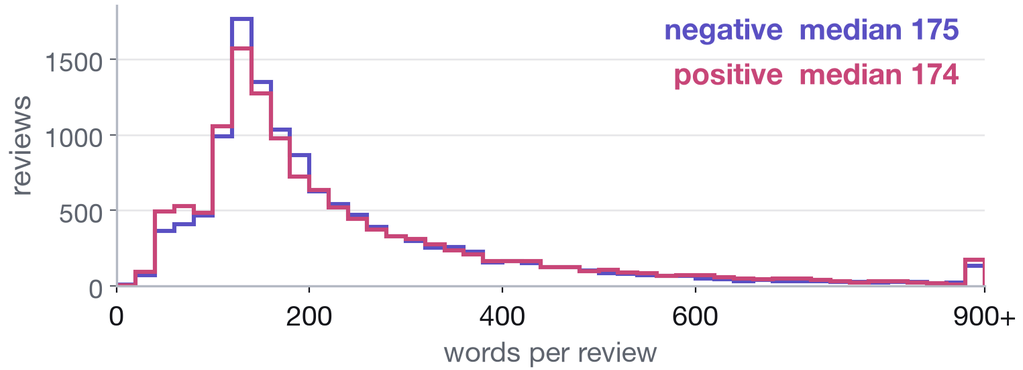
IMDB: balanced movie reviews, judged by accuracy over 15 folds

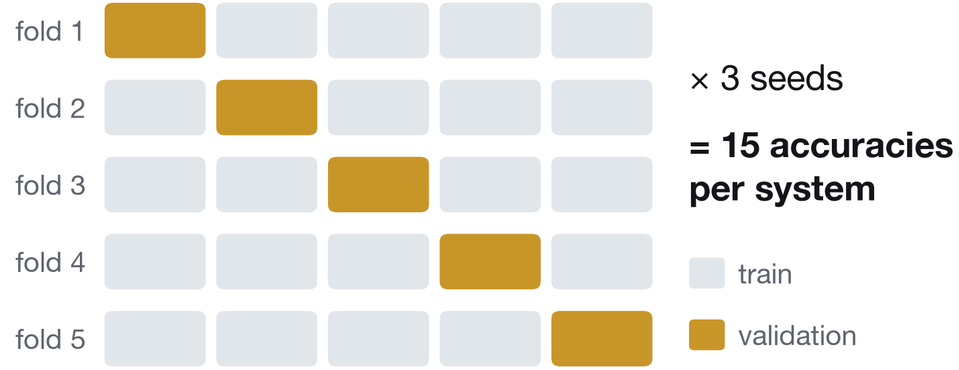
Balanced classes → accuracy is a fair metric (always guessing one class only gets about 50%)



Review length: same for both classes



5-fold cross-validation on the 24,904 training reviews



No preprocessing: case, punctuation and numbers carry sentiment (“NEVER” is more negative than “never”; “10/10” is a rating)

sentiment-classification-bpe-logcounts-mlp

A new BPE tokenizer: 5,000 tokens keep 84% of words whole

How BPE builds tokens: merge the most frequent pair, again and again

training words	cast	part	scary	star	stars
start: letters	c a s t	p a r t	s c a r y	s t a r	s t a r s
merge 1 · in 4 words **a + r → ar**	c a s t	p **ar** t	s c **ar** y	s t **ar**	s t **ar** s
merge 2 · in 3 words **s + t → st**	c a **st**	p ar t	s c ar y	**st** ar	**st** ar s
merge 3 · in 2 words **st + ar → star**	c a st	p ar t	s c ar y	**star**	**star** s

… repeat until the vocabulary is full

Letters-only words that are one token

Vocab size	Words
1,000	64%
2,000	74%
5,000	**84%**
10,000	89%
20,000	94%

Larger vocabulary size

Little difference: more words stay whole, but the vectors get longer and accuracy barely changes.

Larger minimum frequency

No difference: every merged pair already appears hundreds of times, so the tokenizer stays the same.

Training data: 24,904 reviews (5.8M words) are enough for a tokenizer made for movie reviews, not a general-purpose one

sentiment-classification-bpe-logcounts-mlp

Three spaces, each giving 5,000 features per review

Training review #11925 (negative)

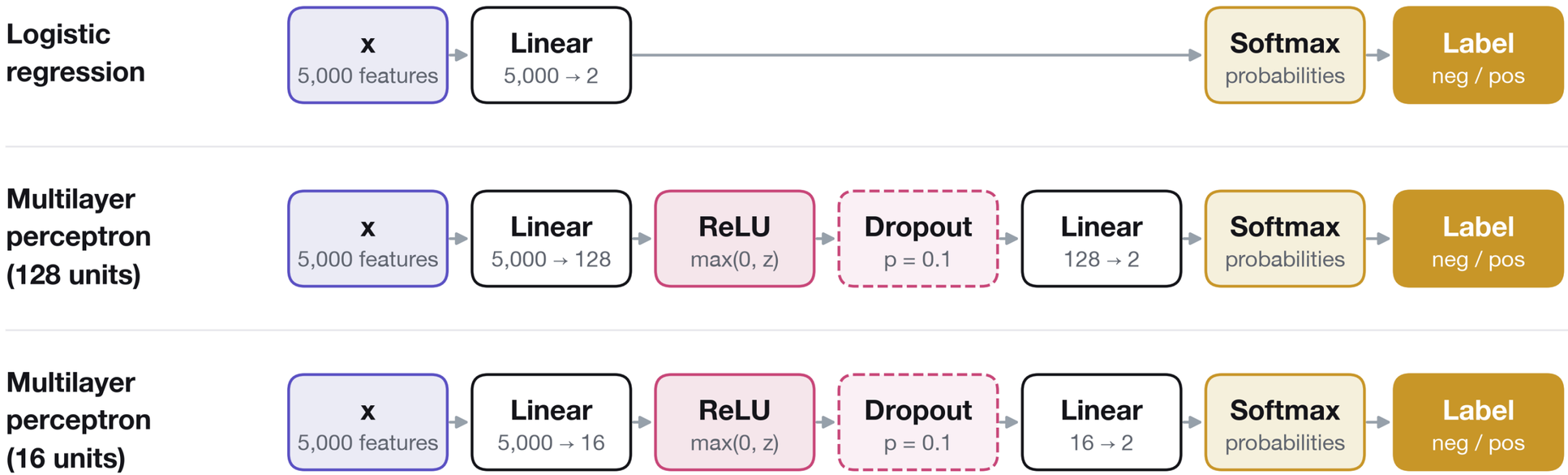
*“This **is** without a doubt the **worst movie** I **have** ever seen. It **is not** funny. It **is not** interesting and should **not have** been made.”*



Why log counts: saying “great” 10 times is more positive than once, but not 10 times more, and frequent words like “the” stay small

sentiment-classification-bpe-logcounts-mlp

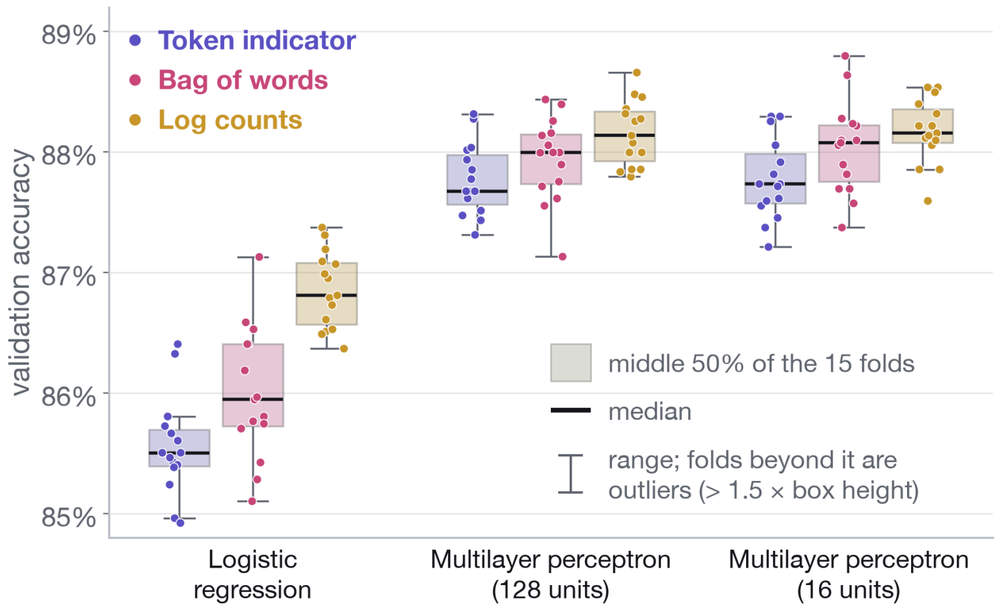
One linear model, two multilayer perceptrons



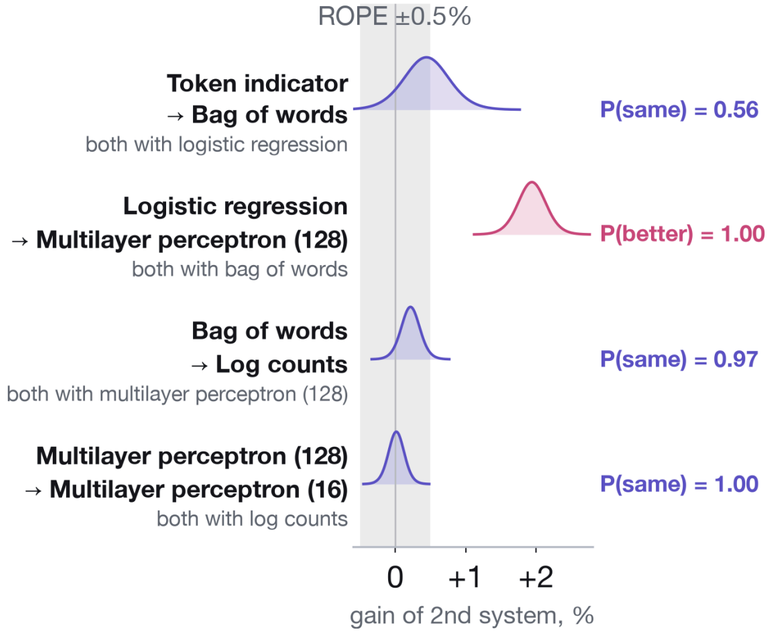
Tried different hyperparameters: minor changes, final setup: learning rate 1e-4, decay 1e-4, dropout 0.1, one hidden layer, 16 units

A hidden layer adds 1–2% accuracy; its size barely matters

Validation accuracy, 15 folds



Bayesian comparison, ROPE 0.5%



Strength and limit: 5-fold cross-validation × 3 seeds gives reliable comparisons, but validation data is not truly unseen

12% of reviews are misclassified; 6.8% by all 15 models

Irony

“OK, this movie is stupid. I mean that in a good way though.”

#24092 · label positive · model 9% pos.

Mixed review

“visually stunning like all of Ang Lee’s work … I was really disappointed”

#1391 · label negative · model 99% pos.

Wrong label

“blown away by just how entertaining this zombie epic is … a cult classic”

#3027 · label negative · model 99% pos.

Fan of bad films

“gloriously ghastly … a delectably dreadful doozy”

#19144 · label positive · model 1% pos.

Verdict at the end

“the weakest film in the original Star Wars trilogy … a winner”

#13279 · label positive · model 1% pos.

Praise, then “but”

“He is a great comedian but this is not his best movie”

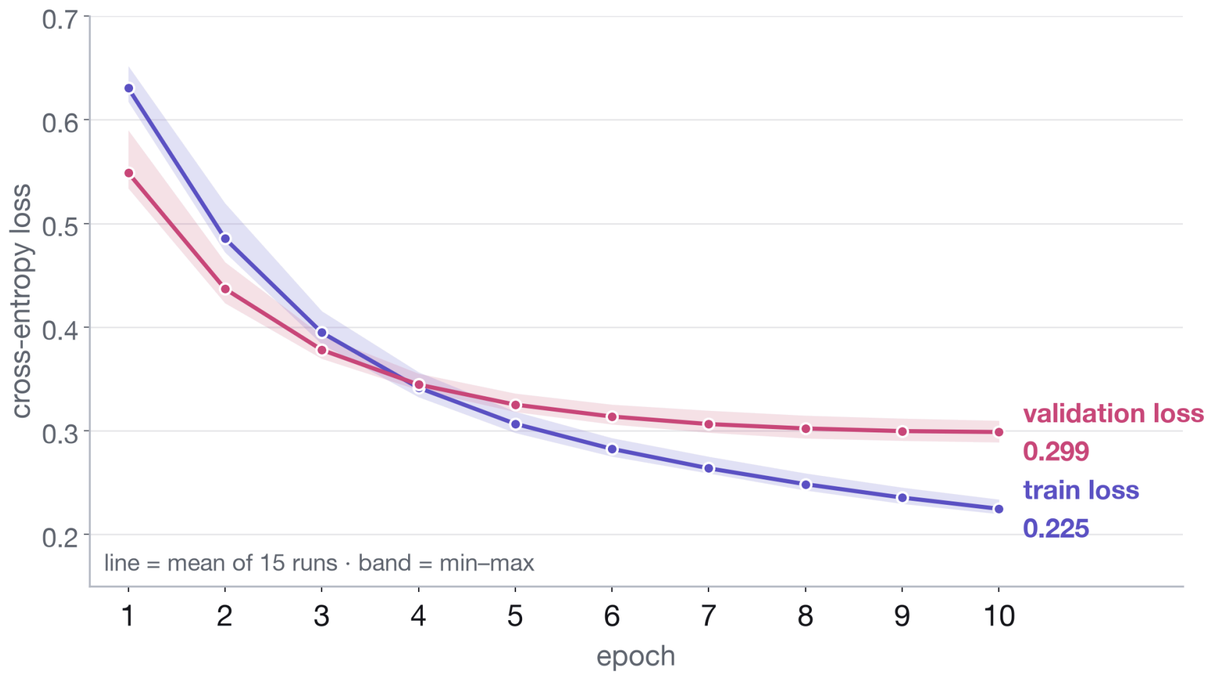
#10931 · label negative · model 94% pos.

Where errors come from: word counts miss context like irony and mixed opinions, and some labels contradict the text

sentiment-classification-bpe-logcounts-mlp

The selected model holds up on unseen test reviews

Learning curves



Test set: 24,678 reviews

87.86%

mean test accuracy of the 15 models, std ± 0.08

88.30% +0.44 pts

ensemble accuracy: majority vote of the 15 models

Validation loss levels off at about 0.30 by epoch 10, while train loss keeps falling: more epochs add little.

Other domains: not directly usable, since its tokens are learned from movie reviews, so it would need a new tokenizer and retraining